ROBERT VOKÁČ, BSc

Software Engineer – C++ Systems & Cross-Platform Development

Prague, Czech Republic · +420 704 780 275 · robertvokac@robertvokac.com

GitHub: github.com/openeggbert · github.com/robertvokac

Web: robertvokac.com · **LinkedIn:** linkedin.com/in/robert-vokáč

PROFILE

Software Engineer with substantial independent work in **modern C++ systems programming**.

I build **3D tooling** (a 3D scene editor and a procedural world generator), **cross-platform frameworks**, compatibility layers, runtime systems, and platform abstractions using C++23, **SDL3, OpenGL**, bgfx, and CMake. My work focuses on portable architecture, API compatibility, rendering/input/audio systems, and application portability across Windows, Linux, WebAssembly, and Android. My public portfolio spans **≈447,000 lines of C++** across the projects below.*

Seeking roles in systems programming, framework engineering, tooling, platform abstraction, or cross-platform software development.

RELEVANT C++ ENGINEERING EXPERIENCE

Independent C++ Systems Developer – OpenEggbert

Open-source engineering work · 2024 – present

Designed and built modern C++ 3D tooling, framework, and compatibility-layer projects focused on portability, API behavior reproduction, and platform abstraction.

Main technologies: **C++20/23, SDL3, OpenGL, bgfx, CMake, Windows, Linux, WebAssembly, Android**

Mesh Craft — 3D Scene Editor (flagship project) ≈28.0k LOC

GitHub: github.com/openeggbert/mesh-craft · **Web:** meshcraft3d.com · meshcraft.openeggbert.com

C++23 desktop 3D scene editor for the XML-based MC3 format — an editable, human-readable scene description compiled to standard runtime formats.

- Designed the `.mc3.xml` scene format: primitives, hierarchical groups, PBR materials, prefabs, layers
- Implemented CSG boolean operations (union/difference/intersection via Manifold) and extrude-along-path
- Built keyframe animation (position, rotation, scale, visibility, material channels) with a timeline editor
- Created glTF/GLB and binary MCB exporters with CLI tools (`mc3togltf`, `mc3tomcb`)
- Built the Dear ImGui editor (orbit camera, gizmos, undo/redo, autosave) on the CNA runtime (SDL3/OpenGL ES 3)

Mesh World — Procedural 3D World Generator ≈16.6k LOC

GitHub: github.com/openeggbert/mesh-world · **Web:** meshworld3d.com · meshworld.openeggbert.com

Procedural 3D world/city generator built on the Mesh Craft ecosystem, with a real-time exploration app.

- Implemented 20 C++ chunk generators and 11 Lua object generators emitting MC3 scenes
- Embedded a Lua 5.4 scripting sandbox (sol2) for content modding with auto-discovered generators
- Built a planetary map subsystem (quadtree LOD 0-18, 52-biome classification, hydrology and mountain-range terrain algorithms)
- Designed SQLite content packs (taxonomy/material registries, generator bundles) and an MC3 validation pipeline
- Built MeshWorldApp, a standalone SDL3/OpenGL explorer streaming chunks around the player
- 1176 tests passing, zero warnings, CI on every push

CNA — Cross-Platform C++ Framework (XNA-style API) ≈138.0k LOC

GitHub: github.com/openeggbert/cna · **Web:** libcna.com · cna.openeggbert.com · **Demo:** speedyblupi.com/SpeedyBlupi2013 (WebAssembly)

Cross-platform C++ framework implementing an XNA-style programming model over SDL3, validated by a working C# XNA game port running natively on desktop, Android, and WebAssembly.

- Designed a cross-platform C++ framework layer over SDL3
- Built application loop, input, rendering, audio, and resource systems
- Grew the pluggable rendering backend layer to 11 backends: SDL_Renderer, OpenGL (EasyGL), Vulkan, and bgfx, plus WebGPU, native Direct3D 9/11/12 (Wine/DXVK-verified), HTML Canvas, an ASCII glyph-grid backend, and a DirectDraw-shaped backend (DX3) fronting the free-direct project
- Ported a C# / XNA application to native C++, replacing .NET, MonoGame, and original XNA dependencies
- Produced stable builds from one codebase for Windows, Linux, WebAssembly, and Android

CNA Samples — XNA Sample Collection Ported to C++ ≈48.0k LOC

GitHub: github.com/openeggbert/cna-samples · **Demo:** web demos coming soon

C++ ports of the official Microsoft XNA Game Studio 4.0 sample collection, running on CNA — a continuously growing compatibility test bed proving the framework's API coverage on real code.

CNA Craft — Voxel World Prototype on CNA ≈6.3k LOC

GitHub: github.com/openeggbert/cna-craft · **Status:** in development

Minecraft-like first-person voxel-world prototype built directly on the CNA framework, a faithful C++ port of fogleman/Craft: unbounded chunk-streamed block terrain with baked ambient occlusion, a 54-block roster, world-editing commands, signs, SQLite delta-based world persistence, and multiplayer against its own server. Runs on CNA's OpenGL (EasyGL), Vulkan, and bgfx backends, and is playable in the browser via WebAssembly/WebGL2.

Sharp Runtime — C#/.NET Runtime Subset in Native C++ ≈70.7k LOC

GitHub: github.com/openeggbert/sharp-runtime · **Web:** sharpruntime.openeggbert.com

Reimplementation of a pragmatic subset of the .NET runtime (`System::*` namespaces) in idiomatic modern C++: exceptions, events/delegates, collections, string/number/date formatting, and IO. Foundation layer for CNA, Mesh Craft, Mesh World, and the game ports.

Free Direct — DirectX 3 (2D) Compatibility Layer ≈4.3k LOC

GitHub: github.com/openeggbert/free-direct · **Web:** freedirect.openeggbert.com · **Demos:** speedyblupi.com/SpeedyEggbert2 · speedyblupi.com/PlanetBlupi

Modern C++ reimplementation of a game-driven DirectDraw / DirectSound subset over SDL3.

- Created DirectDraw-style 2D rendering abstractions: CPU surfaces, blitting, palettes, color keys, locking, presentation
- Reproduced selected DirectX 3 behavior without obsolete SDK dependencies

Free API — Win32 Compatibility Layer ≈4.4k LOC

GitHub: github.com/openeggbert/free-api · **Web:** freeapi.openeggbert.com

Minimal Win32 API (circa 1998) compatibility layer on SDL3 — windowing, message loop, input translation, multimedia timers, and MIDI/MCI music — enabling legacy Windows games to run on Linux, macOS, Android, and the web. Works together with Free Direct as the system/platform side of legacy application compatibility.

Free Eggbert — Speedy Eggbert 2 Reconstruction ≈28.1k LOC

GitHub: github.com/openeggbert/free-eggbert · **Web:** freeeggbert.openeggbert.com · **Demo:** speedyblupi.com/SpeedyEggbert2 (WebAssembly, partial)

Game-preservation project: a buildable reconstruction of the 1998–2001 game Speedy Eggbert 2, reverse-engineered from original binaries (Ghidra, IDA, ILSpy) and made portable beyond Windows/DirectX via Free API + Free Direct, including an Emscripten web build with persistent saves.

Mobile Eggbert — C++ Port of Speedy Blupi (2013) ≈20.5k LOC

GitHub: github.com/openeggbert/mobile-eggbert · **Web:** mobileeggbert.openeggbert.com · **Demo:** speedyblupi.com/SpeedyBlupi2013 (WebAssembly, fully playable)

C++ port of the 2013 Windows Phone XNA game Speedy Blupi, running on CNA. Migration chain: C# XNA 4.0 → ILSpy decompilation → MonoGame → native C++/CNA. Playable on desktop and in the browser (IndexedDB save persistence); the reference implementation for Galaxy Eggbert.

Galaxy Eggbert — 3D Game on the CNA Stack ≈14.3k LOC

GitHub: github.com/openeggbert/galaxy-eggbert · **Web:** galaxyeggbert.openeggbert.com

3D remake of Speedy Blupi / Mobile Eggbert validating the full CNA + Easy3D engine stack; targets Linux, Windows, WebAssembly, and Android from a single C++ codebase.

easy-gl — OpenGL / OpenGL ES RAII Wrapper ≈3.6k LOC

GitHub: github.com/openeggbert/easy-gl · **Web:** easygl.openeggbert.com

Toolkit-independent C++20 RAII wrapper over OpenGL / OpenGL ES — shaders, buffers, VAOs, textures, framebuffers. The host app owns the window and GL context; used by CNA as its OpenGL rendering backend.

easy-3d — 3D Helper Library for CNA ≈1.0k LOC

GitHub: github.com/openeggbert/easy-3d · **Web:** easy3d.openeggbert.com

Small C++23 helper library beside CNA: cameras (orbit/follow), texture atlas, billboard and cube batching, and debug draw — deliberately a library, not an engine. First consumer: Galaxy Eggbert.

meta-gl — Low-Level Type-Safe OpenGL Wrapper ≈7.8k LOC

GitHub: github.com/openeggbert/meta-gl · **Web:** metagl.openeggbert.com

Type-safe C++23 wrapper for OpenGL ES 2.0+ / WebGL: runtime function loading via a host-supplied `GetProcAddress`, `enum class` wrappers for GL constants, 358 thin wrapper functions. Foundation layer beneath easy-gl; Emscripten/WebGL support with context-loss handling.

Mobile Eggbert Legacy — C#/MonoGame Preservation Archive

GitHub: github.com/openeggbert/mobile-eggbert-legacy · **Stack:** C#, XNA 4.0, MonoGame

Preserved C#/MonoGame stage of the Mobile Eggbert migration: ILSpy-decompiled Windows Phone XNA sources with related repositories merged via git subtree.

Mobile Eggbert LibGDX — Java Port

GitHub: github.com/openeggbert/mobile-eggbert-libgdx · **Stack:** Java, LibGDX, Gradle

Java/LibGDX port of Speedy Blupi (Mobile Eggbert) with a small XNA/.NET compatibility bridge; desktop (LWJGL3), Android, and HTML targets.

Sprite Utils — Sprite Utilities and Assets ≈2.2k LOC

GitHub: github.com/openeggbert/sprite-utils

Small C++23 sprite utility library and assets (number spritesheets, web component) supporting the game projects.

YouTube Frontend — Static Index Generator for ArchiveBox ≈2.3k LOC

GitHub: github.com/openeggbert/youtube-frontend · **Web:** youtube.openeggbert.com

C++23 tool that generates static HTML index pages for ArchiveBox video archives, using OpenCV, FFmpeg libraries, and libcurl.

Personal Projects – robertvokac

Hive — Modular Backend Platform ≈45.0k LOC

GitHub: github.com/robertvokac/hive · **Web:** hive.robertvokac.com

Metadata-driven C++23 backend platform demonstrating large-scale application architecture, extensibility, and backend systems design.

- Designed a plugin-based architecture with dependency-aware startup and migrations
- Built a custom model/ORM layer with metadata-driven entity definitions
- Implemented auto-generated REST APIs from shared model metadata
- Developed scheduler, trigger, validation, and automation pipelines
- Added authentication, token management, operational logging, and migration integrity mechanisms
- Created schema-driven frontend generation backed by shared backend metadata

bit-backup — Bit Rot Detection Tool ≈3.1k LOC

GitHub: github.com/robertvokac/bit-backup · **Stack:** C++23, SQLite, OpenSSL

Command-line tool that detects silent data corruption (bit rot) by storing and verifying SHA-512 file checksums in SQLite — parallel hashing, ignore patterns, rotating scrub verification, and CSV reporting for cron integration.

Lexicon — Desktop Knowledge Dictionary ≈2.8k LOC

GitHub: github.com/robertvokac/lexicon · **Web:** lexicon.robertvokac.com · **Stack:** C++23, Qt 6 Widgets, SQLite

Desktop knowledge dictionary for structured technical notes: typed term relationships with backlinks, Markdown content with live preview, full-text search, and light/dark themes.

* Code-line counts measured with cloc (July 2026): C++ sources and headers (`.cpp`/`.hpp`/`.h`), `src/` and `include/` directories only, excluding tests, vendored, and third-party code.

PRODUCTION SOFTWARE EXPERIENCE

Software Engineer – Assist

2017 – present

Production software engineering role focused on backend systems, SQL, legacy codebases, and long-term maintenance.

- Maintained and improved production Java applications and SQL databases
- Diagnosed and resolved production incidents in large legacy applications
- Performed root-cause analysis, debugging, and long-term stability improvements

Technologies: Java, SQL, Git, Bash

CORE SKILLS

Languages: C++20/23, SQL, Java

C++ / Systems: STL, RAII, smart pointers, resource ownership, modular architecture

Cross-platform: Windows, Linux, WebAssembly, Android

Graphics / Platform: SDL3, OpenGL, bgfx, Dear ImGui, DirectDraw-style rendering, platform abstraction

3D / Tooling: scene editing, CSG, glTF/GLB pipelines, procedural generation, Lua scripting (sol2)

Build / Tools: CMake, Git, Bash, CLion, Visual Studio

Backend / Data: SQL, SQLite, REST API design, ORM concepts

Core areas: compatibility layers, legacy behavior reproduction, rendering/input/audio systems

EDUCATION

BSc in Computer Science
Czech University of Life Sciences Prague
2014 – 2017

LANGUAGES

Czech: Native
English: B2